FININSIGHT: Expense Control & Profitability Analytics

Project Report

Submitted in partial fulfillment of the internship/project requirements.

Submitted By

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- **Course:** B.Tech Computer Science & Engineering (E-Commerce Technology)

Tools Used

- Microsoft Excel
- MySQL (phpMyAdmin)
- XAMPP

Project Duration

4 Weeks

Date of Submission

20.07.26

CERTIFICATE

This is to certify that **Anouska MandaL**, a student of **B.Tech Computer Science & Engineering (E-Commerce Technology)**, has successfully completed the project titled **"FinInsight: Expense Control & Profitability Analytics"**. The project was carried out using Microsoft Excel and MySQL as part of the internship/project requirements.

Acknowledgement

I would like to express my sincere gratitude to my project mentor, faculty members, and the organization for providing me with the opportunity to work on the project **"FinInsight: Expense Control & Profitability Analytics."**

This project helped me enhance my knowledge of Microsoft Excel, SQL, financial analysis, and dashboard development. I also gained practical experience in data cleaning, KPI development, variance analysis, profitability analysis, database management, and business reporting.

I would also like to thank everyone who supported and guided me throughout the successful completion of this project.

ABSTRACT

The **FinInsight: Expense Control & Profitability Analytics** project is designed to analyze and monitor an organization's financial performance using **Microsoft Excel** and **MySQL**. The project focuses on comparing budgets with actual expenses, evaluating product profitability, and presenting key financial insights through an interactive dashboard.

The project involves importing and organizing financial datasets, calculating Key Performance Indicators (KPIs), performing variance analysis, and generating profitability reports. SQL was used to create a database, import the datasets, and retrieve financial information through queries. The executive dashboard provides a clear visualization of financial performance, enabling faster analysis and informed decision-making.

Overall, the project demonstrates how data analytics tools can be used to improve financial monitoring, reporting, and business decision-making.

INTRODUCTION

Financial management plays a vital role in helping organizations monitor budgets, control expenses, and improve profitability. An effective financial analysis system enables businesses to compare planned budgets with actual expenditures, evaluate revenue and costs, and make informed decisions based on accurate data.

The **FinInsight: Expense Control & Profitability Analytics** project was developed to analyze financial data using **Microsoft Excel** and **MySQL**. The project integrates multiple datasets, including **Budget**, **Expenses**, and **Revenue**, to calculate key performance indicators (KPIs), perform variance analysis, and measure product profitability.

An interactive dashboard was created in Microsoft Excel to visualize financial performance through charts, KPIs, and summaries. Additionally, SQL was used to store, manage, and analyze the financial data efficiently. This project demonstrates how data analytics techniques can support financial reporting and improve business decision-making.

PROJECT OBJECTIVES

The primary objectives of the **FinInsight: Expense Control & Profitability Analytics** project are:

- To analyze financial data using **Microsoft Excel** and **MySQL**.
- To compare **planned budgets** with **actual expenses**.
- To calculate and monitor **Key Performance Indicators (KPIs)**.
- To evaluate **product-wise profitability** using revenue and cost data.
- To develop an **interactive dashboard** for financial visualization.
- To perform **SQL-based analysis** for efficient data management and reporting.
- To support **data-driven financial decision-making** through meaningful insights.

TOOLS & TECHNOLOGIES USED

The **FinInsight** project was developed using the following tools and technologies:

1. Microsoft Excel

Microsoft Excel was used for data preparation, KPI calculations, variance analysis, profitability analysis, and the development of the interactive dashboard. Various Excel features such as formulas, PivotTables, charts, and conditional formatting were used to analyze and visualize financial data.

2. MySQL (phpMyAdmin)

MySQL, accessed through phpMyAdmin, was used to create the project database, import the financial datasets, and execute SQL queries for data retrieval and analysis.

3. XAMPP

XAMPP provided the local server environment required to run Apache and MySQL, enabling database management through phpMyAdmin.

PROJECT METHODOLOGY

The FinInsight project was completed by following a systematic approach to analyze financial data and generate meaningful business insights.

Step 1: Data Collection

The Budget, Expenses, and Revenue datasets were collected and prepared for analysis.

Step 2: Data Preparation

The datasets were reviewed, organized, and formatted to ensure consistency and accuracy before analysis.

Step 3: KPI Development

Key Performance Indicators (KPIs), including Total Budget, Total Expenses, Total Revenue, and Total Profit, were calculated to provide a summary of financial performance.

Step 4: Financial Analysis

Variance analysis was performed to compare budgeted and actual expenses, while profitability analysis evaluated product performance based on revenue and cost.

Step 5: Dashboard Development

An interactive dashboard was created in Microsoft Excel to visualize KPIs, financial trends, and analytical insights using charts and graphs.

Step 6: SQL Database & Analysis

The datasets were imported into MySQL, and SQL queries were executed to retrieve, summarize, and analyze financial information

DATASET DESCRIPTION

The **FinInsight** project uses three datasets to analyze financial performance and generate business insights.

1. Budget Dataset

The Budget dataset contains the planned budget allocated to different departments for each month. It is used to compare planned spending with actual expenses and identify budget variances.

Fields:

- Department
- Month
- Budget Amount

	A	B	C	D
1	Departme	Month	Budget	
2	Sales	Jan	50000	
3	Marketing	Jan	40000	
4	HR	Jan	15000	
5	IT	Jan	30000	
6	Finance	Jan	20000	
7	Sales	Feb	52000	
8	Marketing	Feb	41000	
9	HR	Feb	15000	
10	IT	Feb	30000	
11	Finance	Feb	20000	
12	Sales	Mar	51000	
13	Marketing	Mar	42000	
14	HR	Mar	16000	
15	IT	Mar	31000	
16	Finance	Mar	21000	
17	Sales	Apr	53000	
18	Marketing	Apr	43000	
19	HR	Apr	16000	
20	IT	Apr	31000	
21	Finance	Apr	21000	
22	Sales	May	54000	
23	Marketing	May	44000	
24	HR	May	17000	
25	IT	May	32000	
26	Finance	May	22000	

2. Expenses Dataset

The Expenses dataset records the actual expenses incurred by different departments. It helps monitor spending patterns and supports variance analysis.

Fields:

- Date
- Department
- Category
- Amount

	A	B	C	D
1	Date	Departme	Category	Amount
2	02-Jan	Sales	Travel	7000
3	03-Jan	Marketin	Advertisi	12000
4	04-Jan	HR	Recruitme	3000
5	05-Jan	IT	Software	8000
6	06-Jan	Finance	Office Su	2000
7	10-Jan	Sales	Client Me	5000
8	12-Jan	Marketin	Social Me	9000
9	15-Jan	IT	Hardware	12000
10	18-Jan	HR	Training	4000
11	20-Jan	Finance	Audit	6000
12	03-Feb	Sales	Travel	6500
13	05-Feb	Marketin	Advertisi	13000
14	07-Feb	HR	Recruitme	3500
15	09-Feb	IT	Software	8500
16	11-Feb	Finance	Office Su	1800
17	15-Feb	Sales	Client Me	5200
18	18-Feb	Marketin	Events	10000
19	20-Feb	IT	Hardware	10000
20	23-Feb	HR	Training	4500
21	25-Feb	Finance	Audit	5500
22	02-Mar	Sales	Travel	7200
23	04-Mar	Marketin	Advertisi	14000
24	06-Mar	HR	Recruitme	3800
25	08-Mar	IT	Software	9000
26	10-Mar	Finance	Office Su	2200
27	14-Mar	Sales	Client Me	4800
28	17-Mar	Marketin	Social Me	9500
29	20-Mar	IT	Hardware	11000

3. Revenue Dataset

The Revenue dataset stores product-wise revenue and cost information. It is used to calculate profit and evaluate product profitability.

Fields:

- Product
- Month
- Revenue
- Cost

	A	B	C	D
1	Product	Month	Revenue	Cost
2	Product A	Jan	120000	70000
3	Product B	Jan	80000	60000
4	Product C	Jan	150000	90000
5	Product D	Jan	60000	55000
6	Product E	Jan	100000	65000
7	Product A	Feb	125000	72000
8	Product B	Feb	82000	61000
9	Product C	Feb	152000	91000
10	Product D	Feb	62000	56000
11	Product E	Feb	102000	66000
12	Product A	Mar	128000	73000
13	Product B	Mar	85000	62000
14	Product C	Mar	155000	93000
15	Product D	Mar	64000	57000
16	Product E	Mar	105000	67000
17	Product A	Apr	130000	74000
18	Product B	Apr	87000	63000
19	Product C	Apr	158000	94000
20	Product D	Apr	65000	58000
21	Product E	Apr	108000	68000
22	Product A	May	132000	75000
23	Product B	May	89000	64000

Purpose of Each Dataset

Dataset	Purpose
Budget	Stores planned budget for each department.
Expenses	Records actual departmental expenses.
Revenue	Stores revenue and cost data for profitability analysis.